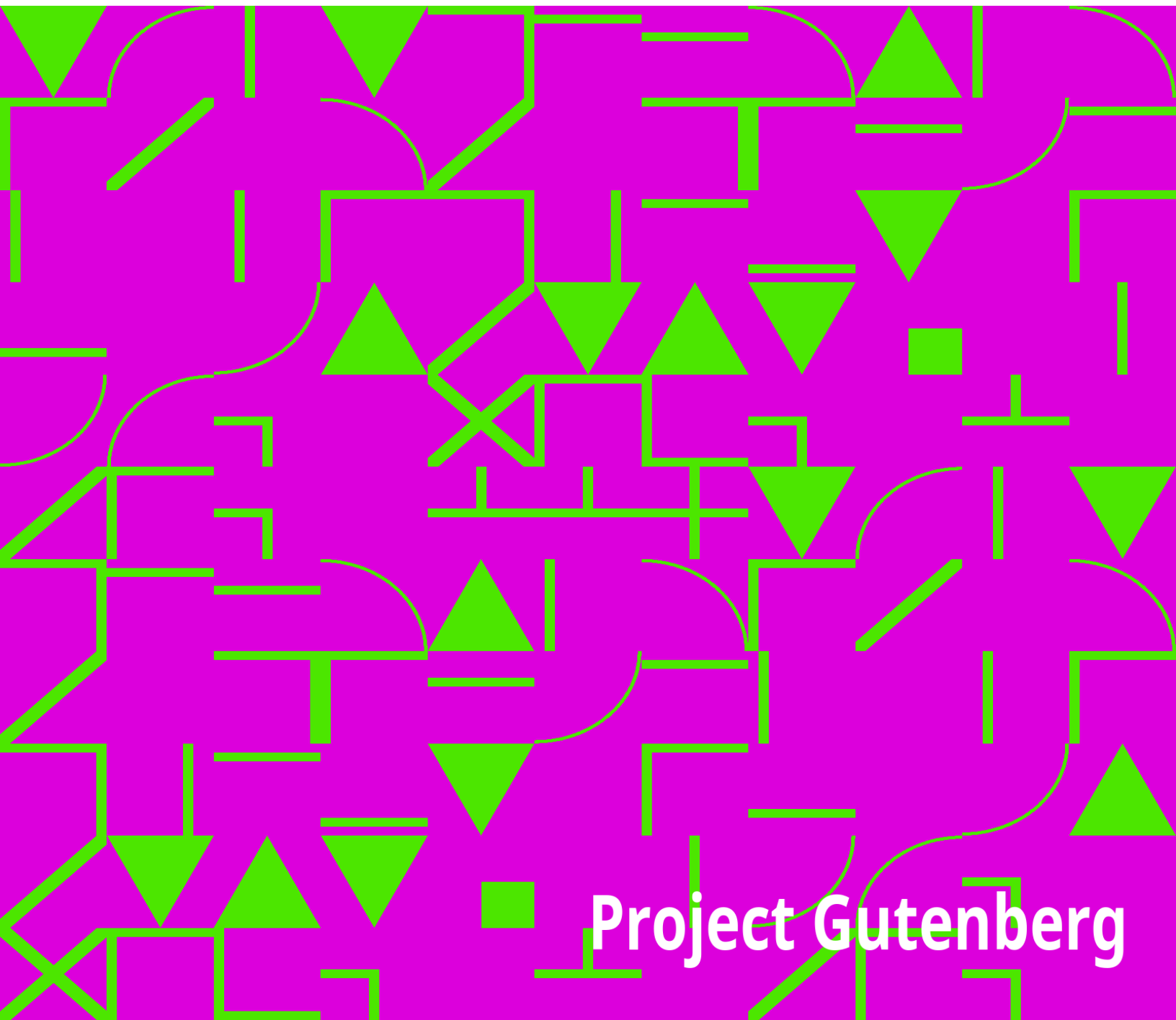


Subspeciation in Pocket Gophers of Kansas, [KU. Vol. 1 No. 11]

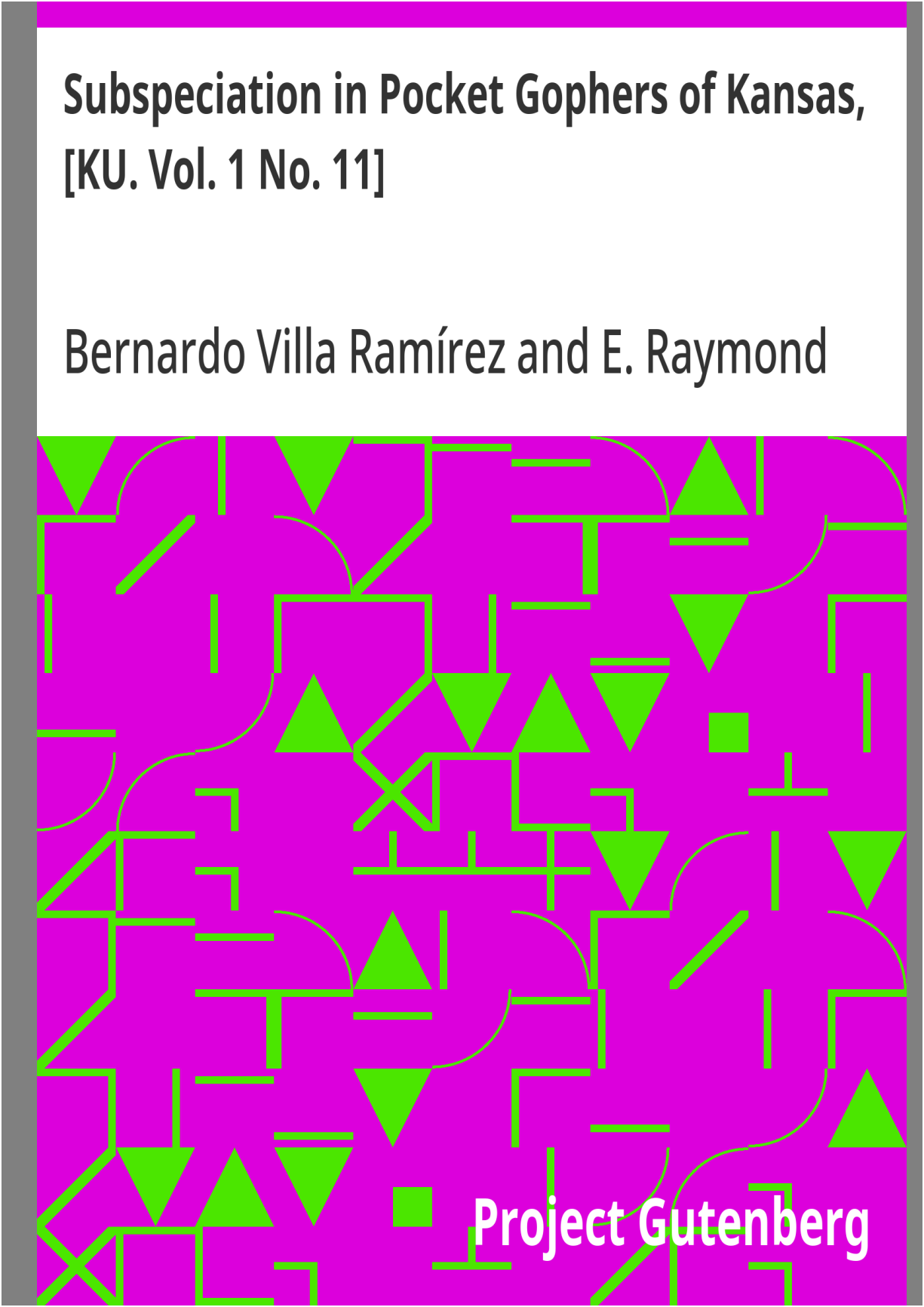
Bernardo Villa Ramírez and E. Raymond

An abstract geometric pattern consisting of various shapes like squares, triangles, and circles, some filled with solid colors (blue, green, yellow) and others outlined in black, set against a white background.

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[KU. Vol. 1 No. 11]**

Bernardo Villa Ramírez and E. Raymond



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By
BERNARDO VILLA-R. and E. RAYMOND HALL

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BERNARDO VILLA-R. AND E. RAYMOND HALL

Several full species of the genus *Geomys* have been recorded from Kansas. The purpose of the study now reported upon was to determine the present taxonomic status of these animals and the distribution of each within the boundaries of Kansas. No pocket gopher of any kind has been reported from the southeastern part of the state; in all other parts *Geomys* is locally common.

HISTORY

The first published reference that we have found to pocket gophers of Kansas is Prof. Spencer F. Baird's (1857:377, 380) mention of two specimens from Fort Riley. One he identified as *Geomys bursarius* (p. 377) and the other (p. 380) he doubtfully referred to *Geomys breviceps*. Both specimens were obtained by Dr. W. A. Hammond. J. A. Allen (1874:49) reported pocket gophers from Kansas under the generic name "Geomys?". Professor M. V. B. Knox (1875:21) published a list of Kansas mammals in which he used the names *Geomys bursarius* Shaw and *Geomys breviceps* Baird, the last one for the specimen taken by Dr. Hammond, at Fort Riley. Baker (1889:57) employed the name *Geomys bursarius* Rich. for the gopher "found along the hundredth meridian, between N latitude 38° 30' and 39° 30'." He reported this animal as common in western Kansas. Merriam (1895:129) recorded *G. bursarius* and *G. lutescens* from Kansas. Allen (1895:265) recorded five specimens of *Geomys lutescens* collected between September 16 and October 13 at Long Island, Phillips County, Kansas, by W. W. Granger. Since that time several papers, some of them dealing mostly with habits of pocket gophers, have been published in which reference is made to *Geomys* in Kansas. Hibbard (1933:240) recognized three species: *G. bursarius*, *G. lutescens*, and *G. breviceps llanensis*. In 1944 (74-75) he recorded *Cratogeomys* from Meade County, on the basis of two skulls dug out of the ground, and he recognized the same three full species of the genus *Geomys* that he did in 1933, along with two additional subspecies.

Specimens to the total number of 335 from Kansas have been available for the present study of the five subspecies recognized. The reason for arranging all of the named kinds as subspecies of a single species is that intergradation has been found to occur between every pair of kinds having contiguous geographic ranges. The characters previously thought by some writers constantly to differentiate, say, *Geomys lutescens* of western Kansas from *Geomys bursarius* of eastern Kansas, prove not to do so; instead, in areas geographically intermediate between the geographic ranges of the two kinds, the pocket gophers are intermediate in morphological characters and therefore are regarded as intergrades. Intergradation of this kind here is accepted as the criterion of subspecies, and lack of such intergradation as the criterion of species. Search for structural characters, distinctive of the different kinds, additional to those characters noted by other writers, has resulted in the finding of a few such characters but they too are subject to intergradation. Therefore the several kinds are arranged as subspecies of a single species which takes the name *Geomys bursarius* because it is the oldest available name. Detailed comment on specimens showing intergradation are to be found in the accounts of *G. b. bursarius* and *G. b. major*.

METHODS AND ACKNOWLEDGEMENTS

The series with the largest number of individuals from one restricted locality was selected for initial study. These individuals were segregated by sex, and specimens of each sex were arranged from oldest to youngest. Each series was divided into age-groups, and within a given age-group of one sex from one locality of what was considered as one species, estimation was made of the amount of individual variation. Thus, it was possible when comparing different kinds of pocket gophers to use only one age class of one season of one sex.

Age was estimated to some extent by size of animal and nature of its pelage. The immature pelage is grayer and the hair is more crinkled than in adults. A more certain guide to age, however, is furnished by the skull. With increasing age some sutures disappear, the rostrum increases in length and the ridges marking the limits of the temporal muscles come to fuse and eventually, in males, form a high sagittal crest.

Cranial measurements were taken as follows:

Basilar length.—From the anteriormost inferior border of the foramen magnum to a line connecting the posteriormost margins of the alveoli of the first upper incisors.

Length of the nasals.—The greatest length of the nasals.

Zygomatic breadth.—The greatest distance across the zygomatic arches.

Mastoid breadth.—The greatest distance across the mastoids.

Breadth of rostrum.—Width, perpendicular to long axis of the skull.

Interorbital constriction.—The least distance between the orbits.

Maxillary tooth row.—The greatest length of the upper molariform tooth row at the alveolar border.

Extension of premaxillae posterior to nasals.—From the posteriormost border of the nasals to the posterior end of the extension of a premaxilla.

Depth of skull.—From the median suture of the frontals, on the dorsal surface of the skull to the median suture of the palatines at the level of the first molar (not premolar).

Length of rostrum.—From the anterior border of the nasal to the maxilla at the lateral end of the hamulus of the lacrimal.

In the list of specimens examined, localities are arranged by counties from west to east, beginning at the northwestern corner of the state; specimens in each county are arranged from north to south. If several localities are in the same latitude, the westernmost is listed first. Capitalized color terms are after Ridgway, Color Standards and Color Nomenclature, Washington, D. C., 1912.

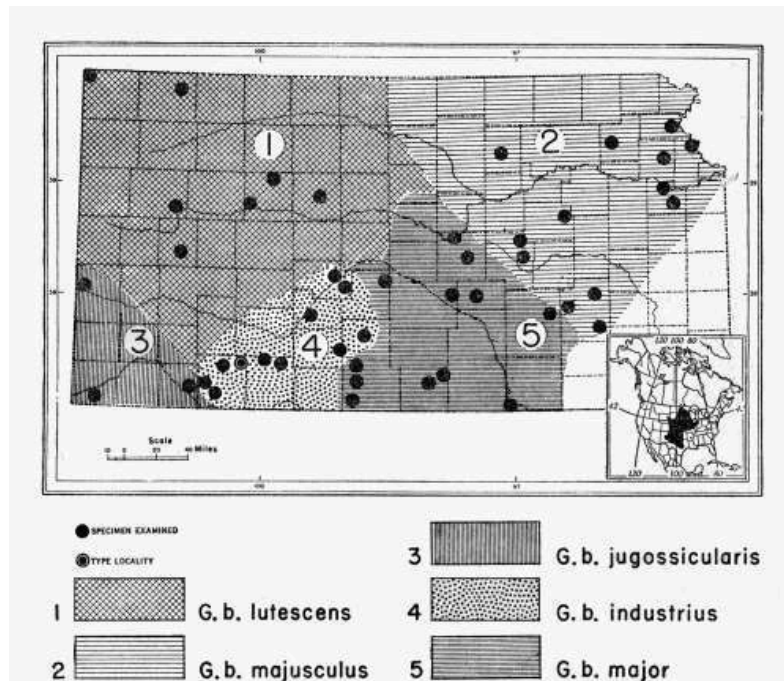


Fig. 1. Map showing the geographic distribution of the five subspecies of the Mississippi Valley pocket gopher, *Geomys bursarius*, in Kansas, with insert showing range of the species.

In connection with this study each of the authors acknowledges assistance from the John Simon Guggenheim Memorial Foundation and one of us (Villa) is grateful for assistance also to Drs. Isaac Ochoterena and Roberto Llamas of the Biological Institute of Mexico. For the loan of specimens we are grateful to Dr. William B. Davis, of the Agricultural and Mechanical College of Texas; Dr. G. C. Rinker, of Hamilton, Kansas; and Mr. A. J. Kirm, of Somerset, Texas. Unless otherwise indicated, specimens are in the University of Kansas Museum of Natural History.

ACCOUNTS OF SUBSPECIES

Geomys bursarius lutescens Merriam

Geomys bursarius lutescens Merriam, North Amer. Fauna, 4:51, October 8, 1890; Scheffer, Technical Bull., U. S. Dept. Agric., 224:6, January, 1931.

Geomys lutescens Merriam, North Amer. Fauna, 8:127-29, January 31, 1895; Lantz, Trans. Kansas Acad. Sci., 19:175, 1905; Lantz, Kansas State Agric. College Bull., 129:335, April, 1905; Hibbard, Trans. Kansas Acad. Sci., 36:240, 1933; Black, 30th Bienn. Rept. Kansas State Board Agric., 35:182, 1937; Swenk, Missouri Valley Fauna, 2:1, February 1, 1940; Allen, Kansas State Teachers College, Emporia, Bull. Inf. in Educ., 20 (no. 5):15, May, 1940; Hooper, Occas. Papers Mus. Zool., Univ. Michigan, 420:3, June 28, 1940.

Geomys lutescens lutescens, Hibbard, Trans. Kansas Acad. Sci., 47:74, 1944.

Type locality.—Sandhills on Birdwood Creek, Lincoln County, western Nebraska.

Distribution in Kansas.—Northwestern Kansas, eastward certainly to Ellis County, southward certainly to Scott County.

Description.—Animals with total length averaging no more than 272 mm.; length of vertebrae of tail averaging no more than 92; hind foot averaging no more than 35. Color: In autumn pelage, upper parts Light Ochraceous-Buff becoming Buckthorn Brown in middorsal region and there forming a faint longitudinal band; sides Pale Yellow Orange. In summer, Buckthorn Brown on upper parts with a dorsal band, especially distinct on specimens from Ellis and Trego counties; specimens from farther west lack the distinct dorsal band. Underparts Gray Drab and sometimes whitish, usually whitish in young specimens; basal color of pelage Deep Neutral Gray; fore and hind feet whitish. Skull: Zygomatic arch broadly and squarely spreading anteriorly; temporal impressions uniting to form a low sagittal crest in adult males, but in adult females and in young males the impressions usually remain apart; shape of interparietal varying from subquadrate in young specimens to subtriangular or triangular in adults; in some young specimens the interparietal is reduced to a minute, ovoid bone.

Comparisons.—See comparisons in the accounts of other subspecies occurring in Kansas.

Remarks.—In his monographic revision of the pocket gophers, Merriam (1895:129) recorded 3 "typical or nearly typical" specimens from Trego County, and 18 "non typical" specimens as follows: Garden Plain, Sedgwick County, 4; Belle Plain, Sumner County, 5; Cairo, Pratt County, 6; Kiowa, Barber County, 2; and Ellis, Ellis County, 1. A detailed discussion of Merriam's account of the distribution of *Geomys lutescens* in Kansas is given by Swenk (1940:11-12).

Judging by specimens in the University of Kansas Museum of Natural History, *G. bursarius lutescens* in Kansas is restricted to the northwestern part of the state, reaching southward certainly to Scott County and eastward certainly to Ellis County; precise limits of distribution of this subspecies are unknown. Additional collecting is necessary to determine where the range of *lutescens* meets the ranges of the other subspecies. The specimens studied are remarkably uniform. One specimen obtained in October, in Trego County, is slightly lighter colored than any other from Kansas. In other characteristics it agrees with specimens from northwestern Kansas and from the type locality.

Specimens examined.—Total number 32, as follows: *Cheyenne County*: 23 mi. (by road) NW St. Francis, 3. *Rawlins County*: 2 mi. NE Ludell, 10. *Logan County*: 5 mi. W Elkader, 3; no locality more precise than county, 1. *Trego County*: Wakeeney, 4; 12 mi. S Collyer, Perrington Ranch, 3; no locality more precise than county, 5. *Scott County*: 4 mi. S Scott City, 2. *Ellis County*: Hays State College Campus, Hays, 1.

Geomys bursarius majusculus Swenk

Geomys bursarius majusculus Swenk, Missouri Valley Fauna, 1:6, December 5, 1939; Hibbard, Trans. Kansas Acad. Sci., 47:74, 1944.

Geomys bursarius, Baird, Expls. and surveys for a railroad route from the Mississippi River to the Pacific Ocean, pt. 1, Mammals, 377, 1857; Merriam, North Amer. Fauna, 8:120, January, 1895; Lantz, Trans. Kansas Acad. Sci., 19:175, 1905; Lantz, Kansas State Agric. College Bull., 129:335, April, 1905; Scheffer, Kansas State Agric. College Ento. and Zool. Dept. Bull., 172:199, September, 1910; Hibbard, Trans. Kansas Acad. Sci., 36:240, 1933; Allen, Kansas State Teachers College Emporia Bull. Inf. Stud. in Educ., 20 (no. 5):15, May, 1940.

Geomys bursarius bursarius, Black, 30th Bienn. Rept. Kansas State Board Agric., 35:181, 1937.

Geomys breviceps, Baird, Expls. and surveys for a railroad route from the Mississippi River to the Pacific Ocean, pt. 1, Mammals, 380, 1857.

Type locality.—Lincoln, Lancaster County, Nebraska.

Distribution in Kansas.—Northeastern Kansas, westward certainly to Clay and Marion counties and southward certainly to Greenwood County.

Description.—Color: Upper parts Mummy Brown in fresh appearing pelage of February but in more worn pelage of March more reddish being near (16') Prout's Brown; top of head and sometimes back darker than rest of upper parts; underparts usually with some whitish anteriorly; fore and hind feet and approximately distal half of tail white. Size: Large, total length averaging more than 280 mm. in males and 257 in females; hind foot averaging 35 mm. or more in males. Skull: Large; rostrum averaging more than twice as long as wide; sagittal crest high in males and barely present in females; occiput vertical when skull is laid top down; least width of braincase less than distance from alveolus of upper incisor to middle of lateral border of P⁴ at alveolar border.

Comparisons.—From *Geomys bursarius lutescens*, *majusculus* differs as follows: Color darker, Mummy Brown to Prout's Brown instead of Buckthorn Brown. In both sexes: head and body a fifth to a sixth longer; hind foot 5 to 6 per cent longer; skull averaging larger in all parts measured except that premaxillae (in each subspecies) extend equally far posteriorly to nasals; diastema longer in relation to basilar length; rostrum longer relative to its width; sagittal crest higher; rostrum often more depressed distally; angle of suture between maxilla and jugal more obtuse.

From *G. b. bursarius*, according to Swenk (1939:6), *majusculus* differs in larger size.

From *G. b. illinoensis*, *majusculus*, according to Komarek and Spencer (1931:405), differs in brownish instead of slate-gray coloration and in two cranial characters as follows: Nasals straight-sided instead of shaped like an hour-glass, and superficial canals on palatine extending anteriorly beyond first molar, and from there anteriorly more or less separated. The first of these characters does not always hold; occasional individuals of *majusculus*, for example some from Douglas County, have the nasals shaped like an hour-glass.

From *G. breviceps dutcheri*, *majusculus* differs in larger size (hind foot more than 33 mm. in males, and 29 in females; basilar length more than 42 mm. in males and 36 in females); dorsal exposure of jugal longer than width of rostrum measured between ventral margins of infraorbital foramina.

From *G. bursarius major* of southcentral Kansas (for example Harvey County), *majusculus* differs in slightly darker color, being Mummy Brown instead of Prout's Brown; size larger (in males total length more than 284 mm., hind foot 35 or more, basilar length of skull more than 42,

and in females total length 265 or more, hind foot averaging 33 or more, and basilar length 40 or more).

Skull: Averaging larger in all parts measured, except that premaxillae do not extend so far posteriorly to nasals in either males or females; interorbital constriction slightly narrower in adult females; temporal ridges forming a more prominent sagittal crest in adult males (sagittal crest barely present in some adult males of *major* from Harper County).

Remarks.—In employing the subspecific name *majusculus* we are following Swenk (1939:6) who on the basis of larger size differentiated the animals from southeastern South Dakota, the eastern parts of Nebraska and Kansas, and the western and southern parts of Iowa, from *G. bursarius bursarius* to which he assigned a more northern geographic range. In the absence of comparative materials of the northern subspecies we cannot make an independent decision on the validity of *majusculus* and recognize that if it is inseparable from *G. b. bursarius* the latter name will apply to specimens from northeastern Kansas. We are the more uncertain about applying the name *majusculus* to specimens from eastern Kansas because they average smaller than topotypes. Only at the northeasternmost locality in Kansas (3 mi. N Cummings, Atchison County) do specimens average as large as topotypes of *majusculus*. Farther southward they become progressively smaller in eastern Kansas, and we interpret this as intergradation with the still smaller subspecies *major*, to the southwest. The average external measurements of two adult males from Atchison County are: 321-99-35. Thirty-six miles farther south, in Douglas County, 16 adult males average 289-80-36. From Hamilton, Greenwood County, 80 miles farther southwest, nine adult males average 284-83-35. The maximum total length recorded at these three localities is: Atchison County, 342 (1 of 2 specimens), Douglas County, 308 (1 of 16 specimens), Greenwood County, 357 (in coll. of Dr. Glenn C. Rinker and 1 of 15 males of all ages involved). It will be seen, therefore, that although there is a trend to smaller average size toward the southward, the maximum of 357 millimeters total length at Hamilton exceeds the maximum of 352 millimeters recorded by Swenk (1939:3) among 86 males at Lincoln where the recorded average is largest.

Four specimens from Salina (Debold Farm) are intermediate structurally, as they are also geographically, between *G. b. majusculus* on the one hand and *Geomys bursarius lutescens* and *Geomys bursarius major* on the other hand. In color they agree with *majusculus*, as they do also in width of nasals posteriorly, in more obtuse angle of the rostrum and maxillary arm of the zygomatic arch. They agree with *G. b. lutescens* in having the occiput inclined anterodorsally, and are intermediate between *majusculus* and *lutescens*, but nearer the latter in size of skull and in length of the rostrum relative to its width.

Specimens examined.—Total number, 148, as follows: *Clay County*: 6 mi. SW Clay Center, 3. *Jackson County*: 10-1/2 mi. WSW Holton, 1; no locality more precise than county, 1. *Atchison County*: 3 mi. N Cummings, 2. *Jefferson County*: Oskaloosa, 1. *Leavenworth County*: Fort Leavenworth (Government Hill, 2; Engineer Hill, 1), 6; no locality more precise than county, 19. *Saline County*: Salina, Debold Farm, 4 (coll. of A. J. Kirm). *Morris County*: 1-1/2 mi. N Council Grove, 3. *Douglas County*: 1 mi. NW Midland, 2; 1 mi. N Lawrence, 1; 2-1/2 mi. W Lawrence, 2; 1 mi. W K. U. Campus, 2; 1 mi. W Lawrence, 2; 1/2 mi. W Lawrence, 2; "W K. U. Campus," 2; K. U. Campus, 4; Lawrence, 23; South Lawrence, 1; 1/2 mi. SW K. U. Campus, 2; Southwest K. U. Campus, 1; Haskell Institute, 1; 4-1/2 mi. S Lawrence, 1; 7 mi. SW Lawrence, 6; 7-1/2 mi. SW Lawrence, 1; 8 mi. SW Lawrence, 1; 10 mi. S Lawrence, 1; 11 mi. SW Lawrence, 3; no locality more precise than county, 15. *Marion County*: 1-1/2 mi. NE Lincolnville, 6; 4 mi. SE Lincolnville, 1; 6 mi. S Lincolnville, 1. *Greenwood County*: Hamilton, 1; 1/2 mi. S Hamilton, 4; 1 mi. S Hamilton, 4; 4 mi. S and 14 mi. W Hamilton, 6; 8 mi. SW Toronto, 1; 8-1/2 mi. SW Toronto, 5; no locality more precise than county, 6.

***Geomys bursarius jugossicularis* Hooper**

Geomys lutescens jugossicularis Hooper, Occas. Papers Mus. Zool., Univ. Michigan, no. 420: 1, June 28, 1940; Hibbard, Trans. Kansas Acad. Sci., vol. 47, p. 75, 1944.

Type locality.—Lamar, Prowers County, Colorado.

Distribution in Kansas.—Extreme southwestern part of state, northward certainly to Hamilton County and south certainly to Morton and Seward counties.

Description.—A yellowish-cinnamon colored animal, with body of medium size, zygomatic plate of maxilla deep and mastoid process small.

Comparisons.—Differs from *Geomys bursarius industrius* in slightly lighter color; occiput not strongly inclined anterodorsally.

From *G. b. lutescens jugossicularis* differs in less buffy coloration and deeper zygomatic plate of maxilla.

Remarks.—*G. bursarius jugossicularis* and *G. bursarius industrius* intergrade in the southern part of Meade County. Some specimens from this area show a coloration resembling that of *G. b. jugossicularis*; nevertheless, one specimen from Morton County has the occiput anterodorsally inclined as in *G. b. industrius*.

Specimens examined from Hamilton County correspond closely to *G. b. jugossicularis*; they agree with it both in color and in cranial characters.

Specimens examined.—Total number, 20, distributed as follows: *Hamilton County*: 1 mi. E Coolidge, Conard Farm, 4. *Morton County*: 12 mi. NE Elkhart, 2; Cimarron River, 12 mi. N Elkhart, 4; no locality more precise than county, 6. *Seward County*: 1 mi. E Arkalon, 4.

***Geomys bursarius industrius*, new subspecies**

Geomys lutescens Merriam, North Amer. Fauna, 8:127, January 31, 1895.

Geomys breviceps llanensis, Hibbard, Trans. Kansas Acad. Sci., 36:240, 1933; Black, 30th Bienn. Rept. Kansas State Board Agric., 35:181. 1937.

Geomys lutescens jugossicularis Hooper, Occas. Papers Mus. Zool., Univ. Michigan, 420:1, June 28, 1940.

Type.—Male, adult, skin and skull, no. 14083 Museum of Natural History, University of Kansas; from 1-1/2 miles north of Fowler, Meade County, Kansas; obtained December 30, 1941, by H. H. Hildebrand, original number 16.

Distribution in Kansas.—Southwestern Kansas from Meade County eastward certainly to Pratt and Clark counties; from Pawnee County southward probably to the Oklahoma boundary.

Diagnosis.—Size of body medium; color of upper parts Cinnamon Brown; skull with occiput strongly inclined anterodorsally in males.

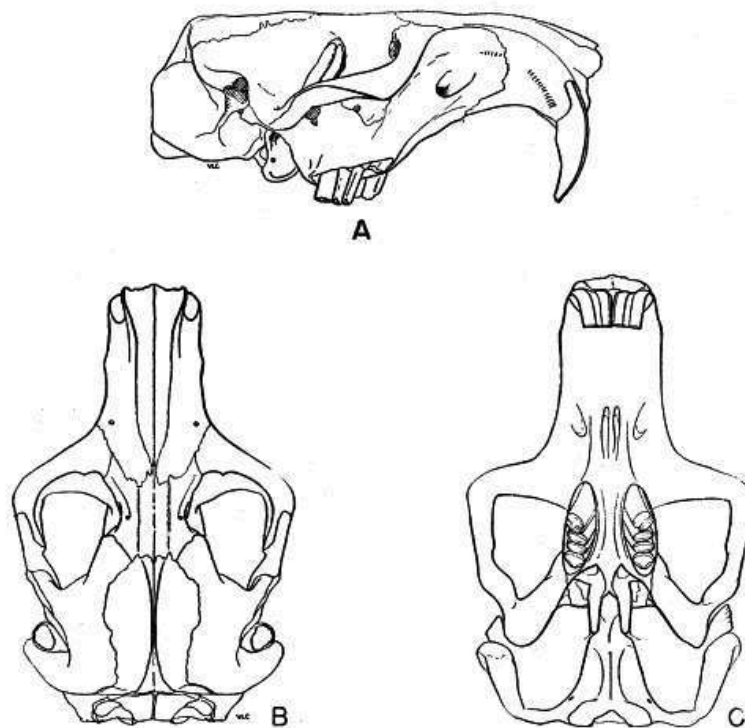


Fig. 2 Three views of the skull of the type specimen of *Geomys bursarius industrius*.
A. Lateral view;

B. Dorsal view;
C. Ventral view.

Description.—Color: Upper parts Cinnamon Brown, slightly reddish, but in some specimens collected in September, in Pawnee County, near (15' i) Ochraceous-Tawny; underparts usually Wood Brown, somewhat whitish anteriorly; forefeet white; hind feet and approximately distal half of tail whitish. Size: Medium (see measurements), total length averaging not more than 271 mm. in males and 254 in females; hind foot averaging not more than 35 mm. in males and less than 32 in females. Skull: In males, least width of braincase equal to distance from alveolus of incisor to anterior border of alveolus of first upper molar, occiput strongly inclined anterodorsally, temporal impressions usually united in a low sagittal crest, zygomatic arch heavy and curved at level of jugal bone. In adult females least width of braincase approximately equal to distance from alveolus of incisor to anterior border of alveolus of first upper molar (not premolar); occiput less inclined anterodorsally than in males; temporal ridges not forming a sagittal crest. In young females the width of the braincase is more than the distance between the alveoli of the incisor and first molar.

Comparisons.—*G. lutescens industrius* differs from *G. lutescens lutescens* in: Color darker; least width of braincase not equal to (usually more than) the distance from the alveolus of incisor to the anterior border of the alveolus of the first upper molar.

G. lutescens industrius differs from *G. lutescens jugossicularis* in: Color slightly darker, the former being Cinnamon Brown instead of Vinaceous Cinnamon, with hairs basally Deep Neutral Gray in upper parts and underparts. Skull: Jugular part of zygomatic arch more curved (convex dorsally) and occiput far more inclined anterodorsally; lower part of mastoidal ridge more prominent.

For comparison with *G. l. major*, see account of that subspecies.

Remarks.—Judging from the known specimens of this subspecies, it has the smallest geographic range of any of the subspecies in Kansas, but additional collecting in Hodgeman County and counties to the north and west of it may extend the known range in those directions; collecting in Comanche County and in adjoining parts of Oklahoma may extend the known range to the southward.

The anterodorsal inclination of the occiput in males is the one cranial character in which *industrius* differs from all of the subspecies with adjoining geographic ranges. The existence of this unique (among adjoining subspecies) cranial character is the principal reason for according subspecific status to this animal. Although it has other characters which are fairly uniform over a considerable geographic area, these other characters, namely, Cinnamon Brown color of the upper parts and medium size of the body, after all, are conditions intermediate between those in *jugossicularis* to the west and those in the darker and larger animals assigned to *major* to the eastward. Considering the intermediate geographic position of *industrius*, the color and size are approximately what a person would predict by study of only the animals to the west and those to the east. Therefore, the color and size probably are indicative of intergradation between *jugossicularis* and *major*. Still, there is the anterodorsally inclined occiput in males—a character of a unique sort—and this influences us to give subspecific status to this animal with full recognition of the fact that it is a "weak" subspecies as compared with any one of the adjoining subspecies.

Hooper (1940:2) in naming as new *Geomys lutescens jugossicularis* referred to his new subspecies a skin-only from Meade County State Park. Our more abundant material from there shows the cranial conformation to be that of *industrius* to which we accordingly assign the specimens. However, with only a skin available, we, too, would have used the name *jugossicularis* because the color is paler than in other specimens of *industrius* and this paleness indicates intergradation between the two named subspecies. Specimens from Pratt County are slightly darker than *industrius* thereby indicating intergradation between *industrius* and *major*.

Specimens examined.—Total number, 58, distributed as follows: *Pawnee County*: Jct. Pawnee and Arkansas rivers, Larned, 6; 1 mi. S and 1 mi. E Larned, 7. *Edwards County*: 1 mi. W and 3-1/2 mi. S Kinsley, 1. *Kiowa County*: Rezeau

Ranch, 5 mi. N Belvidere, 2. *Pratt County*: Pratt, 14; no locality more precise than county, 1. *Meade County*: 3-1/2 mi. NE Fowler, 2; 2 mi. N Fowler, 2; 1-1/2 mi. N Fowler, 2; 1-1/4 mi. N and 3/4 mi. E Fowler, 2; 7 mi. N Meade, Cudahy Ash Pit, 2; 13 mi. SW Meade, 9; State Lake, 2; State Park, 4. *Clark County*: 7 mi. SW Kingsdown, E. A. Stephenson Ranch, 1; 6 mi. S Kingsdown, 1.

***Geomys bursarius major* Davis**

Geomys lutescens major Davis, Texas Agric. Exp. St., Bull. no. 590:32, August, 1940; Hibbard, Trans. Kansas Acad. Sci., 47:75, 1944.

Geomys lutescens Merriam, N. Amer. Fauna, 8:129, January 31, 1895.

Geomys breviceps llanensis, Lantz, Trans. Kansas Acad. Sci., 20 (pt. 2): 215, 1907; Hibbard, Trans. Kansas Acad. Sci., 36:240, 1933; Black, 30th Bienn. Rept. Kansas State Board Agric., 35:182, 1937; Swenk, Missouri Valley Fauna, 2:12, February 1, 1940.

Type locality.—Eight miles west of Clarendon, Donley County, Texas.

Distribution in Kansas.—Southcentral Kansas, northward certainly to Ellsworth County, westward certainly to Stafford and Barber counties and eastward to Cowley County.

Description.—Color: Upper parts varying from Brussels Brown in some specimens to nearly Prout's Brown, especially in specimens from central part of state. Top of head, and sometimes back, darker than rest of upper parts, but no well defined black stripe; underparts varying from whitish to nearly Buffy Brown; fore and hind feet and approximately distal half of tail white. Size: Large (see measurements). Skull: Sagittal crest absent in females and barely present in males; least width of braincase more than distance from alveolus of incisor to middle of lateral border of p^4 at alveolar border. Length of auditory bulla (from anteroventral edge of paroccipital process of exoccipital to hamulus of peterygoid), in each sex, more than 8 mm.; occiput usually vertical when skull is laid top down; zygomatic arch broadly and squarely spreading, divergent anteriorly; rostrum averaging less than twice as long as wide.

Comparisons.—From *G. bursarius lutescens*, *major* differs in color darker, premaxillae extending slightly farther posteriorly; temporal impressions usually forming a more well-marked sagittal crest in males; ventral side of zygomatic arch, at level of jugal bone, more curved.

From *G. bursarius majusculus*, *major* differs in slightly lighter color, smaller size of body; in males, total length less than 284 mm.; hind foot 34 or less; basilar length of skull less than 42; in females total length less than 264, hind foot no more than 33, and basilar length less than 39.

From *G. bursarius industrius*, *major* differs in color, being Prout's Brown, instead of Cinnamon Brown (less Fuscous); body averaging 10 per cent longer; total length in males from 9 to 9.7 per cent longer, hind foot 9.7 per cent longer on the average; skull averaging larger in all parts measured. Occiput less inclined anterodorsally; top nearly flat, less arched than that of *G. b. industrius*; auditory bulla averaging slightly larger and less inflated.

Remarks.—Specimens of this subspecies from Norman, Cleveland County, Oklahoma, and Canton, Dewey County, Oklahoma, and most of those from Kansas, are more Fuscous than topotypes and tend toward *G. bursarius majusculus*. Specimens from McPherson County have a darker dorsal stripe resembling that of *G. bursarius majusculus*. One adult from Little Salt Marsh, Stafford County, is pale, closely resembling topotypes.

Most of the cranial characters, nevertheless, are constant in all available specimens, except that in specimens of each sex from the type locality the basilar length averages 4 to 5 per cent shorter. In the constancy of size of the relatively large auditory bullae and in the nearly flat dorsal profile of the cranial part of the skull, the specimens from Kansas agree with the specimens from the type locality.

Specimens from Harper County have the occiput slightly inclined anterodorsally and thus are reminiscent of *industrius* which has an even greater inclination of the occiput. Probably the appearance in dilute fashion of this character in Harper County is properly to be interpreted as intergradation with *industrius*. If so, the actual intergradation may be to the northwest via Pratt County since specimens from Barber County, immediately west of Harper and lying between Harper County and the range of *industrius*, do not have the occiput so inclined.

Of a pair of adults from eight miles west of Rosalia, Butler County, the female is indistinguishable in color from adults of *G. b. industrius* from northern Meade County and from two specimens from eleven miles west of Clarendon, Donley County, Texas, near the type locality of *G. b. major*. The male from eight miles west of Rosalia is darker as compared either

with *G. b. industrius* or *G. b. major* and the coloration of the upper parts resembles those in *G. b. majusculus*; the underparts are only slightly paler than the upper parts as in *majusculus*. Measurements of the skulls are intermediate between the averages for *G. b. majusculus* and those for *G. b. major*. These specimens from eight miles west of Rosalia are intermediate structurally, and since they are intermediate geographically between *G. b. majusculus* and *G. b. major*, they suggest intergradation of the two subspecies. The specimens in question are referred to *major* because the size is nearer that of *major*. It is mainly the intermediate nature of these two specimens from Butler County, and the intermediate nature of the specimens from McPherson County, Kansas, that have caused us to treat *G. b. majusculus* as only subspecifically distinct from the more western subspecies, *major*.

Specimens examined.—Total number, 77, as follows: *Ellsworth County*: 2 mi. S Ellsworth, 1. *McPherson County*: Smoky Hill River, 1 mi. S and 1/2 mi. W Lindsborg, 5; 1/2 mi. E McPherson, 1. *Stafford County*: Little Salt Marsh, 12; no locality more precise than county, 3. *Reno County*: 8 mi. N and 1 mi. E Haven, 2. *Harvey County*: 1 mi. E and 1/2 mi. N Halstead, 1; Halstead, 3. *Butler County*: 8 mi. W Rosalia, 2. *Barber County*: near South Bridge, Sun City, 1; 2 mi. S Sun City, 1; Wells Ranch, Aetna, 5; "1 mi. W Aetna," 3; near South Bridge, Aetna, 1; near Bridge, 1 mi. S Aetna, 2. *Harper County*: 4-1/2 mi. NE Danville, 8; 1 mi. N Harper, 11; 3 mi. S Harper, 1. *Cowley County*: 3 mi. SW Arkansas City, 4; 3 mi. SE Arkansas City, 9; 3 mi. S Arkansas City, 1.

Measurements of Adult Males of Geomys

(In millimeters)

Key for table headings in table on this page.

N: Number of individuals averaged or catalogue number

L: Total length

T: Length of tail

H: Length of hind foot

B: Basilar length

Na: Length of nasals

Z: Zygomatic breadth

M: Mastoid breadth

Rb: Breadth of rostrum

I: Interorbital constriction

A: Alveolar length of maxillary tooth row

E: Extension of premaxilla posterior to nasals

S: Depth of skull

Rl: Length of rostrum

N	L	T	H	B	Na	Z	M	Rb	I	A	E	S	Rl
<i>G. b. lutescens</i> ; topotypes													
5 ave.	266	82.0	34.2	40.0	17.7	30.5	26.8	11.5	6.7	8.6	3.9	17.1	20.8
min.	257	76.0	33.0	38.3	16.0	29.1	26.1	11.2	6.3	8.1	3.5	16.2	19.1
max.	276	91.0	36.0	42.4	20.3	31.7	27.5	11.9	6.9	9.2	4.2	17.7	23.6
2 mi. NE Ludell, Rawlins Co., Kansas													
12088	272	92.0	35.0	43.2	19.1	32.3	27.7	11.3	6.6	8.4	2.8	18.0	22.1
<i>G. b. majusculus</i> ; Douglas Co., Kansas													
16 ave.	289	79.8	36.3	47.1	21.0	34.1 ^Δ	30.4	12.1	6.8	9.3	3.7	18.5	24.9
min.	273	70.0	32.0	44.7	18.9	30.5	27.5	11.1	6.5	8.2	2.9	17.3	22.9
max.	308	95.0	55.0	49.9	23.2	38.0	34.5	13.5	7.6	10.3	5.7	20.0	28.1
<i>G. b. jugossicularis</i> ; Morton Co., Kansas													
4 ave.	265	82.0	34.2	40.7	16.9	30.0	27.9	10.7	6.0	8.6	5.2	17.3	21.2
min.	250	68.0	30.0	38.5	16.1	29.0	27.5	10.5	5.5	8.2	4.7	16.4	20.2
max.	285	92.0	37.0	42.4	17.4	31.1	28.4	11.0	6.2	9.2	5.5	17.9	22.0
<i>G. b. industrius</i> ; Meade Co., Kansas													
8 ave.	265	82.0	35.0	40.9	18.1	30.0	28.0	11.0	6.2	8.8	4.3	17.7	21.8
min.	247	70.0	33.0	37.9	15.5	28.2	26.5	9.9	5.7	8.0	2.9	16.8	19.5

max.	280	90.0	36.0	43.4	21.0	32.4	29.5	11.6	7.0	9.1	5.2	19.1	24.2
<i>G. b. major</i>; Wells Ranch, Aetna, Barber Co., Kansas													
11724	256	66.0	34.0	41.0	18.3	31.6	28.2	10.6	6.1	9.0	4.0	17.0	21.3
1 mi. W Aetna, Barber Co., Kansas													
11153	240	75.0	32.0	36.7	15.7	26.9	24.6	9.9	5.9	8.8	4.0	15.0	19.5
11152	240	65.0	32.0	36.0	14.2	26.1	25.4	10.9	5.6	8.5	5.0	15.5	18.5
3 mi. SE Arkansas City, Cowley Co., Kansas													
12870	246	76.0	32.0	42.1	16.0 ^E	33.7	29.7	11.5	6.3	9.4	4.5	17.6	21.3
3 mi. SW Arkansas City, Cowley Co., Kansas													
12892	282	84.0	33.0	41.7	17.3		27.7	10.8	6.4	8.9	4.2	17.2	21.5

Measurements of Adult Females of Geomys

(In millimeters)

Key for table headings in table on this page.

N: Number of individuals averaged or catalogue number

L: Total length

T: Length of tail

H: Length of hind foot

B: Basilar length

Na: Length of nasals

Z: Zygomatic breadth

M: Mastoid breadth

Rb: Breadth of rostrum

I: Interorbital constriction

A: Alveolar length of maxillary tooth row

E: Extension of premaxilla posterior to nasals

S: Depth of skull

Rl: Length of rostrum

N	L	T	H	B	Na	Z	M	Rb	I	A	E	S	Rl
<i>G. b. lutescens</i> ; topotypes													
6 ave.	233	72.3	31.1	35.3	15.0	25.9	23.7	10.4	6.1	8.3	3.7	15.4	18.4
min.	215	63.0	30.0	33.5	13.9	24.6	21.8	10.1	5.6	8.1	2.9	14.8	17.3
max.	254	76.0	32.0	37.0	16.8	26.7	24.8	10.7	6.6	8.5	4.5	16.2	19.8
2 mi. NE Ludell, Rawlins Co., Kansas													
11733	230	63.0	31.0	35.3	15.1	26.5	24.1	9.3	6.1	7.5	2.4	15.0	18.2
12155	245	70.0	30.0	35.6	14.6	25.2	24.1	10.6	6.4	7.5	3.1	14.9	18.2
<i>G. b. majusculus</i> ; Douglas Co., Kansas													
17 ave.	265	78.6	32.8	40.6 ^B	17.2 ^B	28.6 ^A	26.4	10.9	6.5	9.1	3.6	16.6	21.0
min.	222	59.0	30.0	37.1	15.9	26.7	24.9	10.0	5.9	8.5	2.0	15.2	18.8
<i>G. b. jugossicularis</i> ; Morton Co., Kansas													
5012	244	72.0	30.0	36.2	16.4	25.4	25.0	10.0	5.9	8.0	4.2	16.0	19.3
5395	230	72.0	30.0	34.6	13.9	24.7	24.8	9.8	5.8	8.0	4.5	15.2	17.5
<i>G. b. industrius</i> ; Meade Co., Kansas													
7 ave.	238	73.0 ^C	31.3	36.4 ^D	14.9	26.3	24.8 ^D	10.0	6.0	8.4	4.1	16.2	18.6
min.	231	65.0	30.0	35.4	14.0	25.8	24.5	9.5	5.6	8.1	3.6	15.5	17.5
max.	256	75.0	32.0	37.8	16.1	27.8	25.9	10.3	6.5	8.7	4.7	17.6	19.9

***G. b. major*; 1 mi. S Aetna, Barber Co., Kansas**

10069	257	95.0	32.0	37.0	16.4	26.4	25.5	10.8	6.2	9.0	3.4	16.4	19.4
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Aetna, Barber Co., Kansas

10070	242	83.0	30.0	36.8	15.7	26.2	25.0	10.1	6.5	9.1	3.3	15.8	19.1
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Wells Ranch, Aetna, Barber Co., Kansas

12238	239	65.0	31.0	34.2	14.5	24.6	23.7	9.6	6.0	8.0	3.6	15.2	17.7
-------	-----	------	------	------	------	------	------	-----	-----	-----	-----	------	------

1 mi. S.Sun City, Barber Co., Kansas

11075	232	66.0	28.0	34.2	14.4	25.0	23.6	9.9	5.9	8.0	3.4	15.0	17.0
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3 mi. SW Arkansas City, Cowley Co., Kansas

12872	242	66.0	30.0	38.1	15.0	28.0	26.2	10.3	6.3	7.8	4.5	16.1	19.1
-------	-----	------	------	------	------	------	------	------	-----	-----	-----	------	------

3 mi. SE Arkansas City, Cowley Co., Kansas

12894	230	82.0	30.0	38.5	15.5	28.0	25.6	10.0	6.7	8.7	4.0	16.6	19.5
-------	-----	------	------	------	------	------	------	------	-----	-----	-----	------	------

12893	246	83.0	32.0	36.5	14.2	25.6	24.8	9.6	6.6	8.7	4.6	15.4	18.1
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Footnotes

[A](#) 15 averaged.

[B](#) 16 averaged.

[C](#) 6 averaged.

[D](#) 5 averaged.

[E](#) approximate.

SUBSPECIES OF THE SPECIES *GEOMYS BURSARIUS*

If *Geomys lutescens major* Davis is correctly judged to intergrade with *Geomys bursarius majusculus* Swenk, the name for the full species will be *Geomys bursarius* because *bursarius* is the oldest name among those available. Some new combinations of names are required. According to our present understanding, the eleven kinds of pocket gophers named below are properly to be arranged as subspecies of the species *Geomys bursarius*:

Geomys bursarius bursarius (Shaw). Type from unknown locality in Upper Mississippi Valley.

Geomys bursarius majusculus Swenk. Type from Lincoln, Lancaster County, Nebraska.

Geomys bursarius hylaeus Blossom. Type from 10 mi. S Chadron, Dawes County, Nebraska.

Geomys bursarius levisagittalis Swenk. Type from Spencer, Boyd County, Nebraska.

Geomys bursarius vinaceus Swenk. Type from Scottsbluff, Scotts Bluff County, Nebraska.

Geomys bursarius lutescens Merriam. Type from Sandhills on Birdwood Creek, Lincoln County, Nebraska.

Geomys bursarius illinoensis Komarek and Spencer. Type from 1 mi. S Momence, Kankakee County, Illinois.

Geomys bursarius jugossicularis Hooper. Type from Lamar, Prowers County, Colorado.

Geomys bursarius industrius new subspecies. Type from 1-1/2 mi. N Fowler, Meade County, Kansas.

Geomys bursarius major Davis. Type from 8 mi. W Clarendon, Donley County, Texas.

Geomys bursarius llanensis Bailey. Type from Llano, Llano County, Texas.

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